

Project #15: MULTI-MODAL MOBILITY MORPHOBOT (M4)

Autonomous Ground-Air Robot for Disaster Response and Search-and-Rescue

Project Type	Primary Focus	Target Environment
Integrated mechatronic design	Morphing mobility and autonomy	Complex disaster zones

Project Overview

The Multi-Modal Mobility Morphobot (M4) is a compact robotic platform capable of changing its locomotion configuration according to terrain, mission urgency, safety and available energy. Unlike conventional robots limited to either driving or flying, M4 combines morphing wheel-track modules, articulated body configurations and integrated propellers. It can roll efficiently on clear ground, crawl or crouch through rubble and confined passages, balance on slopes or obstacles, and transition to aerial flight when gaps, debris or unstable surfaces prevent safe ground access. The project integrates mechanical design, embedded electronics, sensing, real-time control, perception, localization, autonomous navigation, intelligent mode selection and operator supervision.



Figure 1. M4 concept and prototype showing integrated wheel-track structures and aerial propulsion hardware.

Project Objectives and Core Capabilities

- Design a lightweight, robust morphing chassis with protected propellers and modular wheel-track assemblies.
- Develop rolling, crawling, crouching and balancing modes for rubble, slopes and narrow passages.
- Implement stabilized aerial locomotion for obstacle bypass, reconnaissance and inaccessible areas.
- Fuse IMU, LiDAR/depth camera, RGB camera, encoders and proximity sensors for localization and terrain understanding.
- Develop path planning and terrain-aware locomotion-mode selection using safety, time and energy criteria.
- Provide live telemetry, mission logging, remote override and emergency-stop functions.

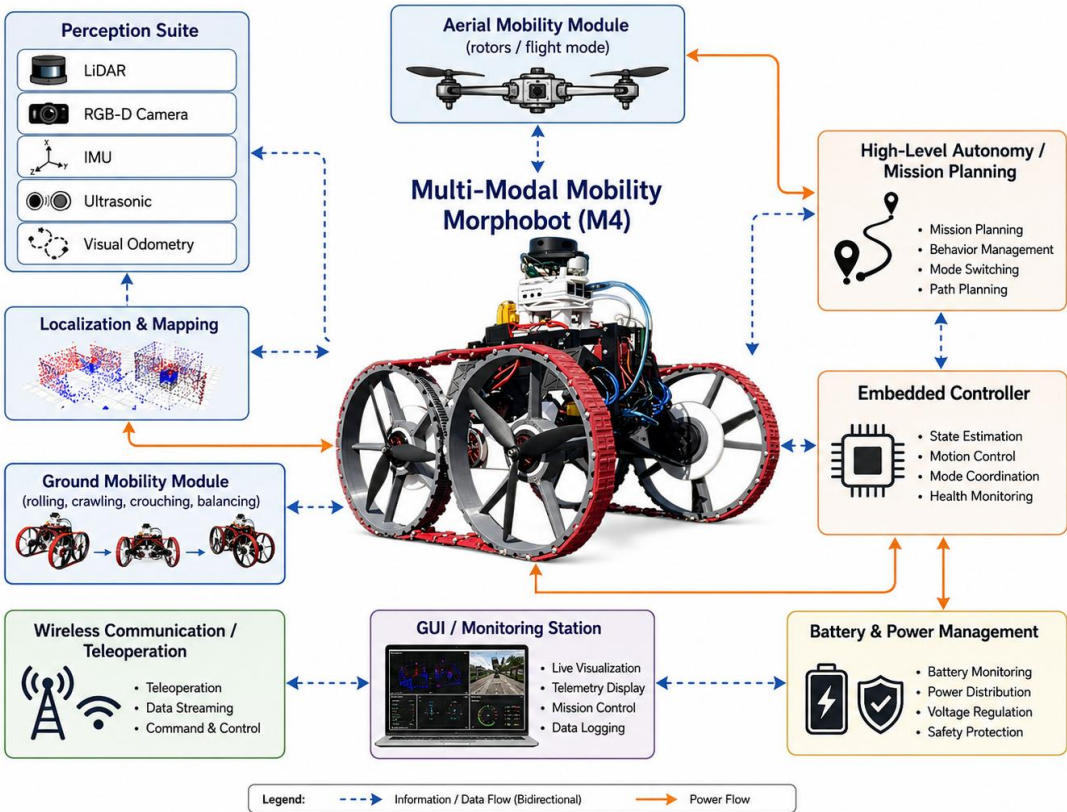


Figure: Overall system layout of the Multi-Modal Mobility Morphobot (M4).

Figure 2. Overall M4 system layout and information/power flow.

SYSTEM DESIGN, HARDWARE AND SOFTWARE ARCHITECTURE

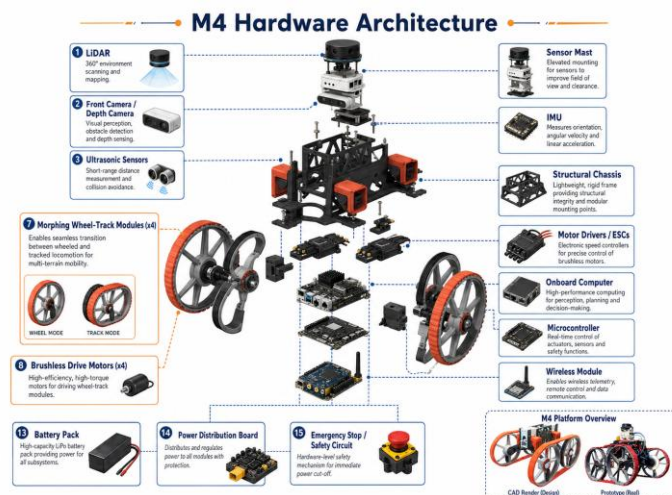


Figure 3. Main mechanical, sensing, computing, actuation, power and safety hardware.

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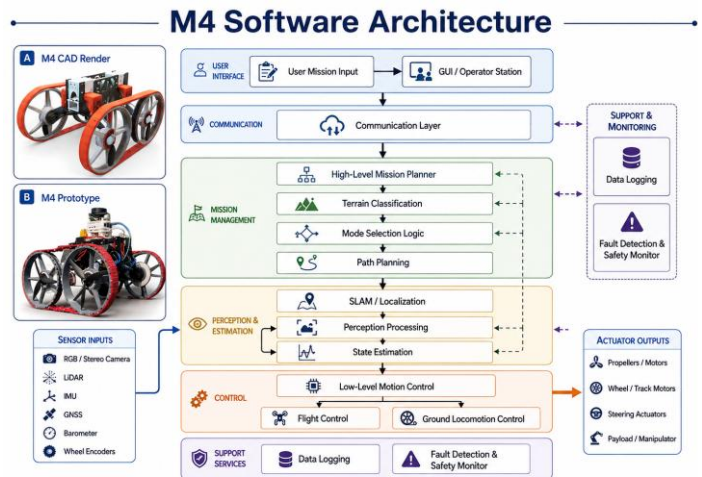


Figure 4. Layered software architecture for autonomy, perception, planning, and control.

Figure 4. Layered software architecture from mission input to actuator outputs.

Technical Development Scope

Mechanical and Actuation

Optimize structural strength, mass distribution, propeller protection, center-of-gravity placement and safe transitions between ground and aerial configurations. Select motors, transmissions, propellers, servos and power electronics using load, thrust and energy calculations.

Autonomy and Control

Develop attitude stabilization, velocity and traction control, path following, obstacle avoidance and a supervisory finite-state machine. The terrain-aware decision layer re-plans and changes locomotion mode when the route becomes unsafe or inefficient.

Sensing and Localization

Integrate IMU, wheel encoders, LiDAR/depth camera, RGB camera and short-range sensors. Implement sensor calibration, visual-inertial odometry and/or LiDAR SLAM for indoor, outdoor and GNSS-denied operation.

Communication and Safety

Provide telemetry, live video, battery and temperature monitoring, data logging, fault detection, watchdog functions, manual override and hardware emergency stop. Safety constraints are enforced before and during every transition.

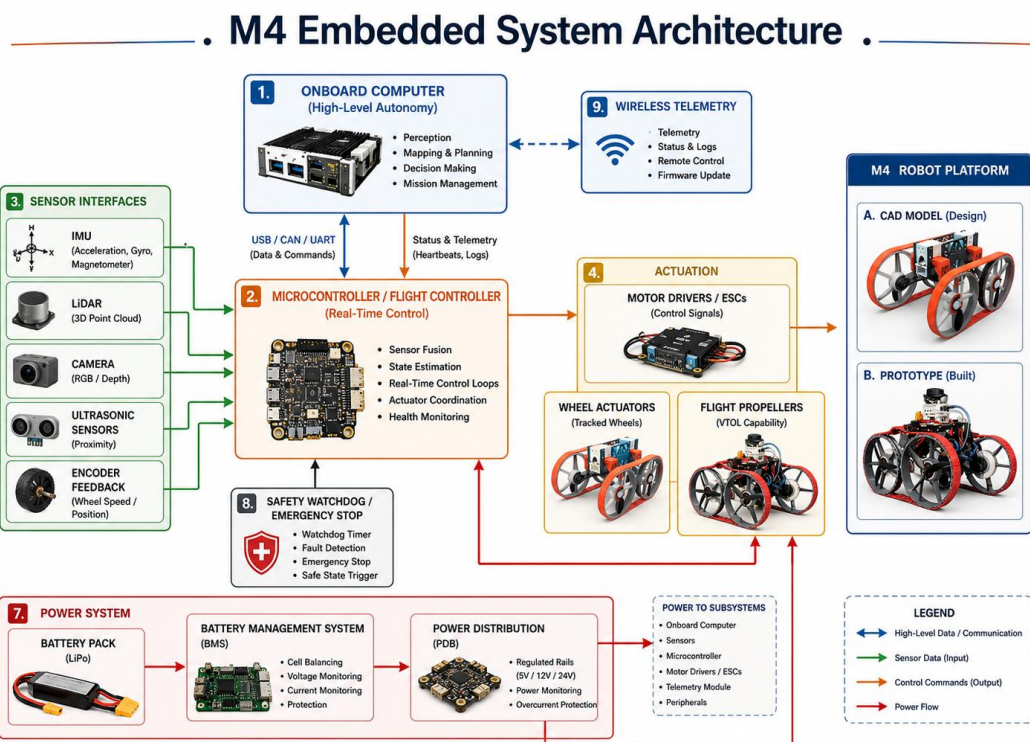


Figure 5. Embedded electronics, sensing, actuation, communication, and power architecture.

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INTELLIGENT MODE SWITCHING, OPERATOR INTERFACE AND VALIDATION

M4 Control Switching and Locomotion Mode Selection

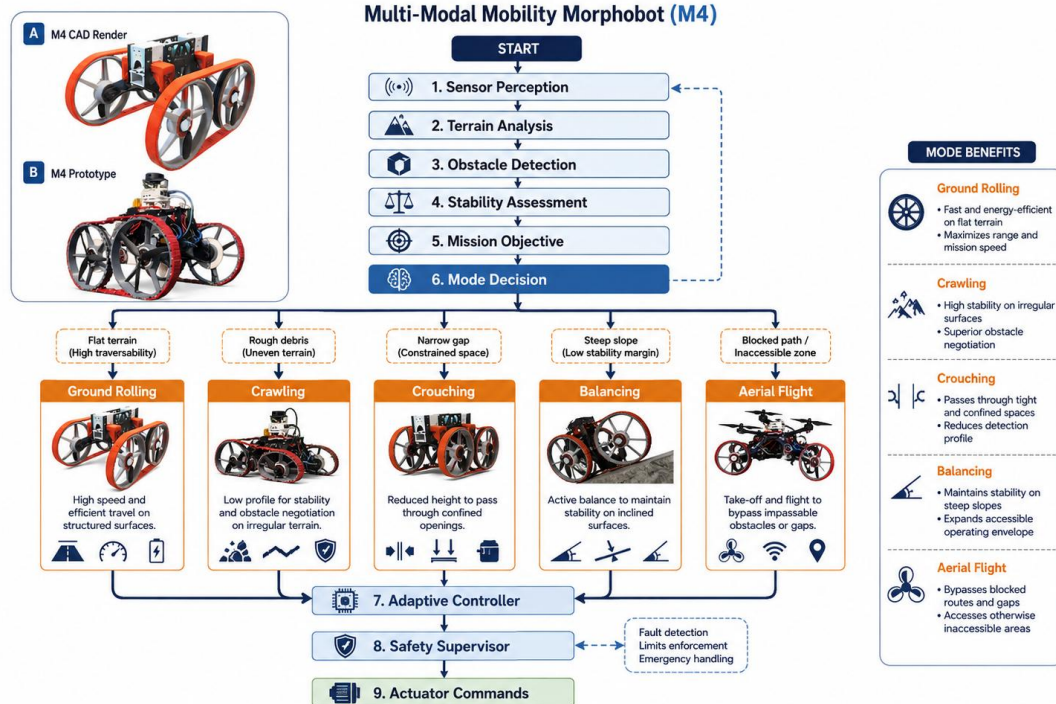


Figure: Control switching pipeline of the Multi-Modal Mobility Morphobot (M4). The robot perceives the environment, evaluates mission and stability constraints, and selects the optimal locomotion mode. All modes are coordinated by the adaptive controller and supervised for safe and reliable operation.

Figure 6. Terrain-aware control switching between rolling, crawling, crouching, balancing and aerial flight.



Figure: Example graphical user interface for commanding and supervising M4.

Figure 7. Mission-control GUI for command, mapping, mode selection and telemetry.



Figure: Example monitoring dashboard for mission supervision, diagnostics, and performance tracking.

Figure 8. Monitoring dashboard for mission status, health, alerts and performance tracking.

Expected Deliverables

- Functional M4 prototype with reliable ground-air transformation and modular mechanical/electrical integration.
- Embedded firmware and autonomy software for sensing, mapping, navigation, adaptive control and mode management.
- Operator GUI with live map, video, health indicators, manual override and mission-data recording.

Prototype Validation Plan and Key Performance Indicators

- Ground mobility: slope, rubble, turning radius and confined-space tests.
- Mode transition and aerial mobility: transition time, repeatability, hover stability, obstacle bypass and endurance.
- Navigation and mission success: localization error, map quality, completion time, energy use and recovery from faults.

Similar concept video: <https://www.youtube.com/shorts/dgVvxrPwB0M>

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